AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM275
Parallel and Distributed Systems

Final Examination

Spring Semester 2024/2025

Date: 2/June/2025

Number of Exam Pages: 2 (including this cover sheet)	Time Allowed: 2 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **5 questions (each question 10 marks)**.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1: Distributed Systems (10 Marks)

a) Define a distributed system and explain three key motivations for using distributed systems in modern applications. [5 marks]

b) Explain the following types of transparency in distributed systems: Access, Location, Failure, Replication, and Concurrency. [5 marks]

Question 2: Cassandra Architecture & Replication (10 Marks)

a) Describe two key architectural features of Apache Cassandra that make it suitable for distributed environments. [4 marks]

b) Suppose you want to create a keyspace for a Cassandra cluster that includes two data centers named `east` and `west`. [6 marks]
- What replication strategy should you use?
- Write the CQL command to create the keyspace using this strategy with replication factors of 2 (east) and 3 (west).

Question 3: Cassandra & Replication (10 Marks)

a) Provide Docker commands to launch two Cassandra nodes in a cluster. [5 marks]

b) Differentiate SimpleStrategy and NetworkTopologyStrategy with usage examples. [5 marks]

Comparison + Code

Question 4: CQL Data Modeling (10 Marks) Code

a) Create a keyspace (university) and a student table using CQL. [5 marks]

b) Using query-first modeling, design a table to get employees who drive a specific car make, sorted by salary descending. Mention keys. [5 marks]

Question 5: AWS Global Service Design (10 Marks)

You are asked to design an AWS-based cloud solution for an international university with students in Asia and Europe. The goal is to ensure low latency, high availability, and secure data access.

Write a detailed plan that includes the following:
- The AWS services and infrastructure components you would use.
- How you would deploy them across Regions and Availability Zones.
- How you would ensure data access is secure and scalable

End of questions

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